

Reactor experiment demonstrates alternative fusion scheme

Startup says result highlights value of plentiful proton-boron fuel

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With its Norman reactor, TAE Technologies aims to demonstrate the value of proton-boron fusion fuel. TAE TECHNOLOGIES

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Researchers in Japan have demonstrated reactions, for the first time in a fusion reactor, with a type of fuel that is plentiful and doesn’t produce damaging particles. Although the reactions were nowhere close to achieving net energy and required even higher temperatures than standard fusion fuel, the result is a proof of principle for private fusion startup TAE Technologies, which argues that its path to a practical power plant faces fewer engineering roadblocks than conventional approaches.

The results show how the alternative fuel, a mix of protons and the element boron, “has a place in utility-scale fusion power,” TAE CEO Michl Binderbauer said in [a statement](#). Not everyone is convinced. “It’s an interesting experiment” but will do little to convince skeptics to switch fuels, says Dennis Whyte, director of the Plasma Science and Fusion Center at the Massachusetts Institute of Technology. 

Fusion is often promoted as a carbon-free energy source that has a plentiful and cheap fuel—a mix of the hydrogen isotopes deuterium and tritium (D-T). In reality, tritium is rare and must be “bred” from lithium in the reactor itself; some scientists are concerned about [future shortages](#). Moreover, when fused at high temperatures, D-T fuel produces copious high-energy neutrons, which are damaging to humans and reactor structures alike.

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TAE is following a different recipe: fusing hydrogen nuclei—protons—with easily mined boron. The reaction generates no neutrons and produces only harmless helium, but it requires temperatures of about 3 billion degrees Celsius—200 times the heat of the Sun’s core and 30 times hotter than what’s needed to fuse D-T. Researchers have already shown they can fuse protons and boron by using particle beams aimed at a solid target or by blasting plasma with lasers. Now, a team has done it—on a small scale, at least—using a conventional fusion reactor, called the Large Helical Device (LHD), at Japan’s National Institute for Fusion Science. The group reported its work last week in *Nature Communications*.

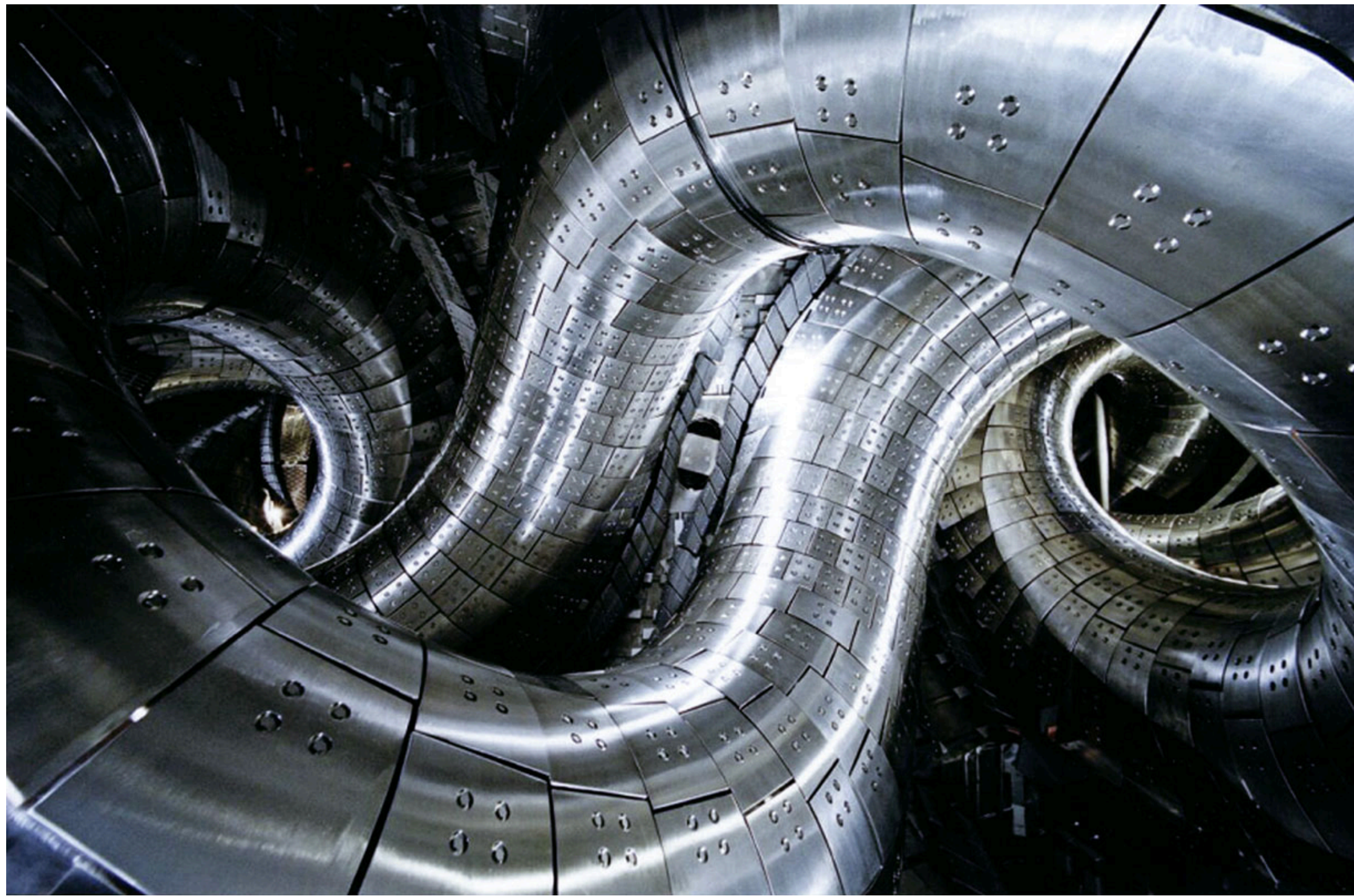
The LHD, which began operations in 1998, is shaped like a twisted doughnut and has electromagnets that contain the superhot ionized fuel, known as plasma. This type of device, known as a [stellarator](#), is not designed to operate at the temperatures required for proton-boron fusion. In the experiments, a boron plasma was heated to 20 million degrees Celsius or so and beams of neutral hydrogen atoms were fired into the plasma. Proton-boron fusion produces high-speed helium atoms, and helium sensors, developed by TAE, [registered 150 times more hits](#) with a boron plasma in the machine than when it contained a nonreactive gas—a sign that fusion was occurring.

Computer simulations from the team suggested this translated into about 5 trillion fusion reactions per second. Although this may sound like a lot, Whyte says it equates to about 7 watts of power, one-tenth of what a candle flame produces. Moreover, Whyte says, most of those reactions were caused by the particle beams. In many fusion reactors, particle beams are used to get the overall plasma temperature hot enough to fuse more widely. But the LHD results suggest fusion was only happening at the few hot spots where the beams hit the plasma, not elsewhere, Whyte says, because the fusion rate drops off rapidly as soon as the beam is turned off.

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A power-producing fusion reactor would need a wider fusion burn to provide enough heat to sustain the reactions—plus some extra to be harvested for electricity. The LHD is a long way from that, but TAE believes it can get there with a very different plasma device. TAE’s various testbeds have created a rapidly spinning “smoke ring” of plasma that is stabilized and heated with particle beams. TAE’s biggest machine so far, called Norman, [achieved a temperature of 60 million degrees Celsius for 30 milliseconds](#).





Researchers say trillions of proton-boron fusion reactions occurred every second within the twisted intestines of Japan's Large Helical Device. NATIONAL INSTITUTE FOR FUSION SCIENCE, JAPAN/SCIENCE SOURCE

In a few years, TAE says it will finish building a successor, called Copernicus, which is intended to reach 100 million degrees Celsius—the temperature needed for conventional D-T fusion. By next decade, the company wants to build an even more powerful machine—Da Vinci—that could take it close to proton-boron temperatures.

A reactor running on protons and boron would remove many of the challenges engineers face as they try to move fusion from scientific demonstration to practical electricity generator. The U.S. National Ignition Facility made headlines last year after demonstrating “gain”: a fusion reaction sparked by powerful lasers that produced more heat than the lasers pumped in. That explosive form of fusion reactor [may be hard to turn into a power plant](#), however. The international ITER reactor under construction in France aims to demonstrate a more steady-state, furnacelike approach. But it won't demonstrate gain until late next decade—when some scientists worry it will begin to gobble up most of the world's supply of tritium.

ITER also has thick concrete shielding to protect operators from neutrons. In a commercial reactor, running round the clock, those neutrons would also damage the reactor's structure and cut short its working life. Studies are underway to find neutron-hard materials for reactors, but no obvious candidates have yet been identified.

Whyte says neutrons are a huge challenge for conventional fusion, but he thinks getting plasma to temperatures measured in the billions could be just as difficult. Even if TAE gets there, each proton-boron reaction yields only one-half of the energy from melding deuterium and tritium. To make it worthwhile, proton-boron fusion would “need strong engineering advantages,” Whyte says.

Correction, 1 March, 2:25 p.m.: A previous version of this story misstated the energy yield of a proton-boron reaction. It is one-half that of a deuterium-tritium reaction, not one-tenth.

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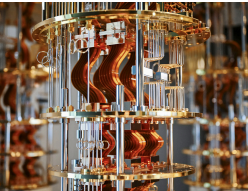
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